

Earthquake Damage Prediction Using Artificial Intelligence Model

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Abstract

Earthquakes are a natural disaster that causes damage to personnel and property. While personnel protection and safety is a well planned response, so can property safety be. This paper presents a LightGBM based machine learning classification model that predicts building damage grades using key features such as floor count, age, plinth area, roof type, and earthquake intensity measures for the Country of Nepal.

Based on the predicted Earthquake Damage grade for a building, measures can be taken to either structurally improve it or the model can be used to predict the damage that would be taken by a building project that is yet to be developed, thereby giving an approximate view in to the future, so that the design can be perfected for Earthquake dense areas.

This model is based on the Nepal 2015 Earthquake Data of the buildings and areas, but can be extended to any Earthquake prone country if the relevant data is collected and given to the model for training.

The model achieves 90.37% overall accuracy and a weighted F1 score of 0.904, performing well across all damage grades. A user friendly GUI allows real time predictions, offering actionable insights for engineers, planners, and emergency responders, enabling faster and more informed post earthquake decision making.

Keywords

Earthquake, Damage Prediction, Machine Learning, LightGBM, Risk Assessment, Nepal Earthquake prediction

Introduction

Earthquakes cause severe damage to buildings, posing risks to life and infrastructure. Rapid damage assessment is critical for effective disaster response but manual inspections are slow and error prone.

Machine learning offers a faster, automated approach by predicting building damage from structural and site-specific features. This study presents a LightGBM based model trained on Nepal earthquake data, using 13 key features and derived metrics.

The model achieves high accuracy and F1 scores across damage grades.

<i>Damage Grade</i>	<i>Inference</i>
0	No damage Building remains fully functional, minor cosmetic issues at most.
1	Minor damage Slight structural or non-structural damage, safe for occupancy with minimal repairs.
2	Moderate damage Visible cracks, partial structural damage, may require careful inspection before use.
3	Extensive damage Significant structural damage, unsafe for occupancy without major repairs.
4	Complete damage Building is collapsed or nearly collapsed, unsafe and likely irreparable.

Related Work

Earthquake damage prediction has traditionally relied on post event surveys, which are slow and error prone. Recent studies use machine learning models like Random Forests, XGBoost, and LightGBM with structural, geographical, and seismic features, achieving faster and more accurate assessments.

This work extends prior research by incorporating derived metrics such as intensity ratios and floor heights and providing a real time prediction interface for practical disaster management.

Methodology

The proposed system predicts building damage grades (0-4) using a LightGBM model trained on Nepal earthquake ward-level data.

 csv_building_structure	Microsoft Excel Comma S...	8,901 KB	No	1,55,369 KB
 mun_level_pred_intensity	Microsoft Excel Comma S...	3 KB	No	8 KB
 ward_level_pred_intensity	Microsoft Excel Comma S...	24 KB	No	54 KB
 ward_vdcmun_district_name_mapp...	Microsoft Excel Comma S...	4 KB	No	51 KB

SAMPLE SNAPSHOTS from Dataset:

csv_building_structure.csv

building_id	district_id	vdcmun_ic	ward_id	count_floc	count_floc	age_buildi	plnth_are	height_ft_	height_ft_	land_surfa	foundation	roof_type	ground_floc	other_floc	position	plan_conf	has_super	has_super	has_super	has_super	has_super	has_super	has
1.2E+11	12	1207	120703	1	1	9	288	9	9	Flat	Other	Bamboo/T Mud	Not applic	Not attach	Rectanguli	0	1	0	0	0	0	0	
1.2E+11	12	1207	120703	1	1	15	364	9	9	Flat	Other	Bamboo/T Mud	Not applic	Not attach	Rectanguli	0	1	0	0	0	0	0	
1.2E+11	12	1207	120703	1	1	20	384	9	9	Flat	Other	Bamboo/T Mud	Not applic	Not attach	Rectanguli	0	1	0	0	0	0	0	
1.2E+11	12	1207	120703	1	1	20	312	9	9	Flat	Other	Bamboo/T Mud	Not applic	Not attach	Rectanguli	0	1	0	0	0	0	0	
1.2E+11	12	1207	120703	1	1	30	308	9	9	Flat	Other	Bamboo/T Mud	Not applic	Not attach	Rectanguli	0	1	0	0	0	0	0	

mun_level_pred_intensity.csv

	Municipality	lat	lng	pred_intensity
0	Champade	27.27953	86.26714	6.947919
1	Chishankh	26.58986	87.21452	5.272554
2	Khiji Demb	27.45878	86.3601	7.169129
3	Likhu Rura	27.86607	85.40144	7.814319
4	Manebhan	27.16987	86.43004	6.542592
5	Molung Ru	27.36621	86.42235	6.854199

ward_level_pred_intensity.csv

ward_id	latitude	longitude	pred_intensity
120101	27.27572	86.27513	6.928824
120102	27.26199	86.23594	6.986425
120103	27.30083	86.27583	6.960487
120104	27.31081	86.34518	6.881559
120105	27.29357	86.32786	6.878566

ward_vdcmun_district_name_mapping.csv

ward_id	vdcmun_id	vdcmun_n	district_id	district_name
120101	1201	Champade	12	Okhaldhunga
120102	1201	Champade	12	Okhaldhunga
120103	1201	Champade	12	Okhaldhunga
120104	1201	Champade	12	Okhaldhunga

Thirteen key building and site-specific features including floor count, age, plinth area, roof type, and seismic intensity are used.

Derived features such as intensity ratio, height estimations, and ward-level damage mean are added to improve predictive performance.

Feature Variable	Type	Meaning
district_id_x	int	Code for the district where the building is located.
vdcmun_id_x	int	Code for the municipality or village.
ward_id	int	Ward number within the district/municipality.
count_floors_pre_eq	int	Number of floors the building had before the earthquake.
count_floors_post_eq	int	Number of floors remaining after the earthquake (if known).
age_building	int	Age of the building in years.
plinth_area_sq_ft	float	Total floor area of the building in square feet.
height_ft_pre_eq	float	Estimated building height before the earthquake.
height_ft_post_eq	float	Estimated building height after the earthquake.
land_surface_condition	categorical	Slope of the land: flat, moderate, or steep.
foundation_type	categorical	Material used for the building's foundation.
roof_type	categorical	Type of roof: thatched, concrete, or metal.
ground_floor_type	categorical	Material of the ground floor: mud, cement, timber, tile, or other.
other_floor_type	int	Flag for any other floor type not listed above.

Feature Variable	Type	Meaning
position	categorical	Whether the building is detached, semi-detached, next to others, or at a corner.
plan_configuration	categorical	Shape or layout of the building (coded 0-9).
has_superstructure_adobe_mud	binary	1 if walls are made of mud bricks/adobe, else 0.
has_superstructure_mud_mortar_stone	binary	1 if stone walls held with mud mortar, else 0.
has_superstructure_stone_flag	binary	1 if stone slab walls, else 0.
has_superstructure_cement_mortar_stone	binary	1 if stone walls held with cement, else 0.
has_superstructure_mud_mortar_brick	binary	1 if brick walls with mud mortar, else 0.
has_superstructure_cement_mortar_brick	binary	1 if brick walls with cement, else 0.
has_superstructure_timber	binary	1 if wooden walls or frames, else 0.
has_superstructure_bamboo	binary	1 if bamboo structures, else 0.
has_superstructure_rc_non_engineered	binary	1 if reinforced concrete without engineering standards, else 0.
has_superstructure_rc_engineered	binary	1 if properly engineered reinforced concrete, else 0.
has_superstructure_other	binary	1 if any other type of wall material, else 0.
condition_post_eq	binary	1 if the building's condition after the earthquake is noted, else 0.

Feature Variable	Type	Meaning
technical_solution_proposed	binary	1 if repair/reinforcement is suggested, else 0.
vdcmun_id_y	int	Municipality code derived from ward info.
vdcmun_name	int	Encoded name of the municipality.
district_id_y	int	District code derived from ward info.
district_name	int	Encoded name of the district.
pred_intensity	float	Predicted earthquake strength at the building's location.
pred_intensity_mun	float	Average predicted intensity for the municipality.
intensity_ratio	float	How strong the earthquake was at this building compared to the municipality average.
avg_intensity	float	Overall average earthquake intensity.
ward_damage_mean	float	Historical average damage in this ward.

The dataset is processed with label encoding for categorical variables and missing value imputation.

The model is trained with an 80/20 train-test split.

A graphical user interface (GUI) allows users to input feature values and obtain real-time damage grade predictions, making the system practical for engineers and disaster responders.

Earthquake Damage Prediction

Earthquake Damage Prediction Tool

District Id X

0

Vdcmun Id X

0

Ward Id

0

Count Floors Pre Eq

0

Count Floors Post Eq

0

Age Building

0

Plinth Area Sq Ft

0.0

Height Ft Pre Eq

0.0

Height Ft Post Eq

0.0

Land Surface Condition

0 - Flat

Foundation Type

0 - Pubble

Results and Discussion

The LightGBM model achieved an overall accuracy of **90.37%** on the test set. Detailed evaluation using precision, recall, and F1 score demonstrates strong performance across all damage grades:

✓ Accuracy: 0.9037

📊 Detailed Classification Report:

	precision	recall	f1-score	support
0	0.9626	0.9164	0.9390	15763
1	0.8281	0.8444	0.8362	17451
2	0.8185	0.7968	0.8075	27283
3	0.8482	0.9175	0.8815	36769
4	0.9963	0.9627	0.9792	55153
accuracy			0.9037	152419
macro avg	0.8908	0.8876	0.8887	152419
weighted avg	0.9060	0.9037	0.9044	152419

📈 Summary Metrics:

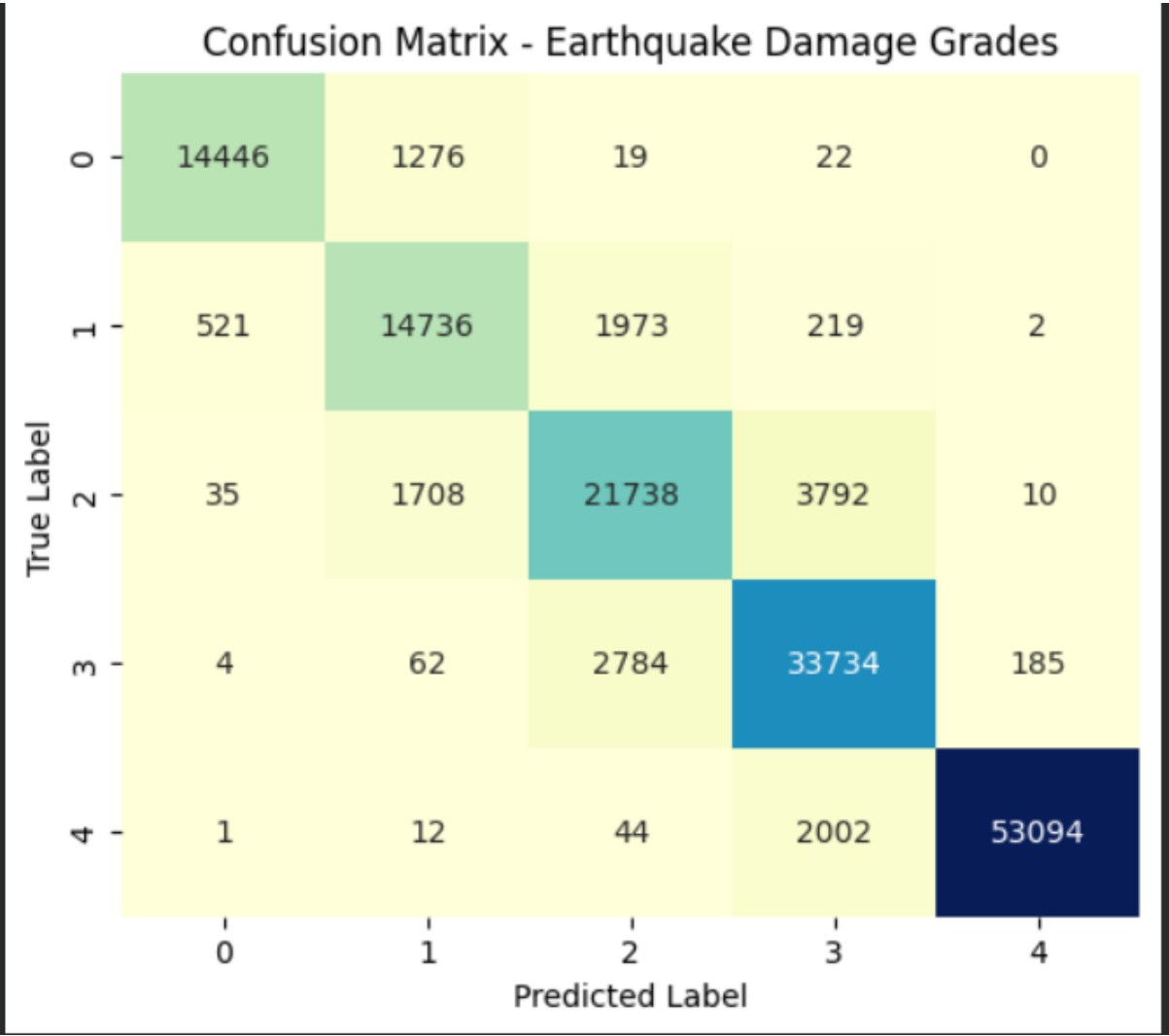
	Metric	Score
0	Accuracy	0.903746
1	Precision (Macro)	0.890765
2	Recall (Macro)	0.887551
3	F1 (Macro)	0.888670
4	Precision (Weighted)	0.906025
5	Recall (Weighted)	0.903746
6	F1 (Weighted)	0.904360

Summary;

- **Grade 0 (No damage):** F1-score 0.9390
- **Grade 1 (Minor damage):** F1-score 0.8362
- **Grade 2 (Moderate damage):** F1-score 0.8075
- **Grade 3 (Extensive damage):** F1-score 0.8815
- **Grade 4 (Complete damage):** F1-score 0.9792

The weighted F1 score of 0.9044 indicates the model’s robustness in handling class imbalance.

Incorporating derived features such as intensity ratios, floor height estimations, and ward level damage means improved predictive performance, especially for moderate and extensive damage grades.



The GUI tool allows for real time damage assessment, enabling engineers, urban planners, and emergency responders to quickly evaluate structural vulnerability

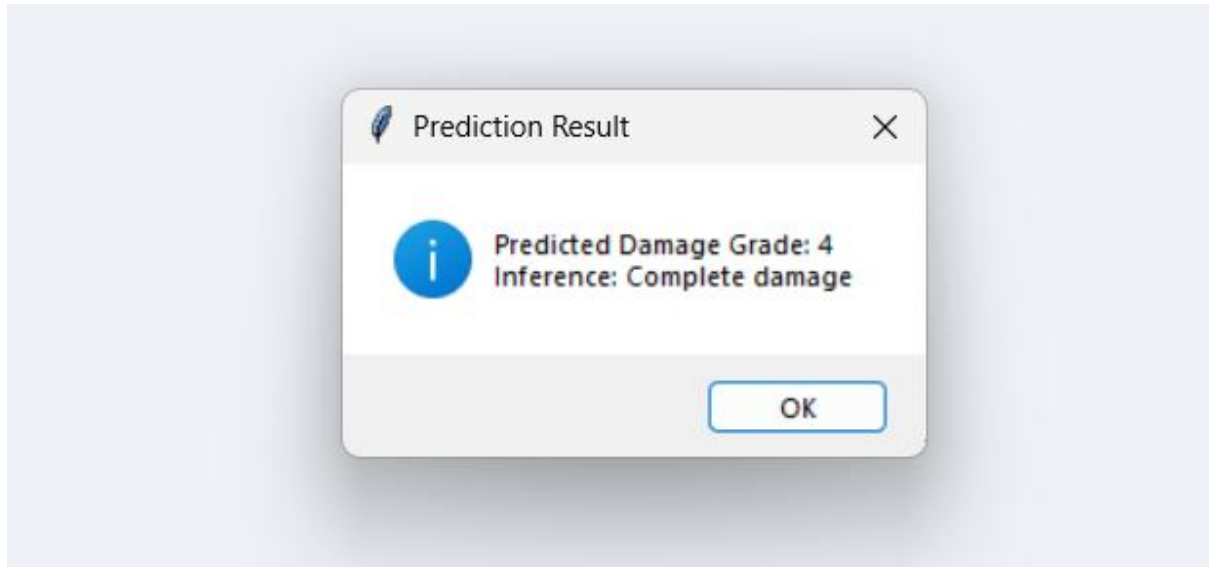
 **Earthquake Damage Prediction Tool**

District Id X	12
Vdcmun Id X	1207
Ward Id	120703
Count Floors Pre Eq	3
Count Floors Post Eq	1
Age Building	15
Plinth Area Sq Ft	500
Height Ft Pre Eq	15
Height Ft Post Eq	5
Land Surface Condition	0 - Flat
Foundation Type	4 - Other
Roof Type	1 - Reinforced concrete
Ground Floor Type	2 - Timber
Other Floor Type	0
Position	3 - Corner
Plan Configuration	3
Has Superstructure Adobe Mud	☐
Has Superstructure Mud Mortar Stone	☒
Has Superstructure Stone Flag	☐
Has Superstructure Cement Mortar Stone	☐

- | | |
|--|-------------------------------------|
| Has Superstructure Cement Mortar Stone | ☐ |
| Has Superstructure Mud Mortar Brick | ☐ |
| Has Superstructure Cement Mortar Brick | ☐ |
| Has Superstructure Timber | ☐ |
| Has Superstructure Bamboo | ☐ |
| Has Superstructure Rc Non Engineered | ☐ |
| Has Superstructure Rc Engineered | ☐ |
| Has Superstructure Other | ☐ |
| Condition Post Eq | ☒ |
| Technical Solution Proposed | ☒ |

Vdcmun Id Y	1207
Vdcmun Name	Champadevi Rural Munic
District Id Y	0
District Name	Okhaldhunga
Pred Intensity	6.9
Pred Intensity Mun	6.3
Intensity Ratio	2
Avg Intensity	6.4
Ward Damage Mean	0.0

Predict Damage Grade



Compared to traditional post event surveys, the system offers faster, automated, and consistent predictions, facilitating informed decision-making and potentially reducing human and economic losses in post-earthquake scenarios.

Conclusion

This study presents a machine learning-based system for rapid earthquake damage assessment of buildings. Using a LightGBM model trained on Nepal ward-level data, the system predicts damage grades (0-4) with high accuracy (90.37%) and a weighted F1 score of 0.9044.

By incorporating both buildings specific features and derived metrics such as intensity ratios and floor height estimations, the model effectively captures structural vulnerability.

The user friendly GUI enables real time predictions, bridging the gap between models and practical disaster management applications.

This approach can significantly aid engineers, urban planners, and emergency responders in prioritizing interventions, ultimately contributing to safer, more resilient communities.

References

[1] “Nepal 2015: Building Damage & Seismic Intensity,” Kaggle Dataset, Sirorororo, “Nepal-Earthquake-Ward-Level-Intensities-and-Impact”, 2019. [Online]. Available: <https://www.kaggle.com/datasets/sirorororo/nepal-earthquake-ward-level-intensities-and-impact>.

[2] S. Sirorororo, “Earthquake Damage Grade Classification,” Kaggle Notebook, 2024. [Online]. Available: <https://www.kaggle.com/code/sirorororo/earthquake-damage-grade-classification>.